

Exercise: Data Visualization – California Housing Dataset Analysis

Name: Haorong Guo (郭昊荣)

Student ID: 12411217

1. Task 1: Feature Distribution Visualization with Kernel Density Estimation

1.1 KDE of Features

- KDE With Original Data (kernel width set by Scott's Rule)

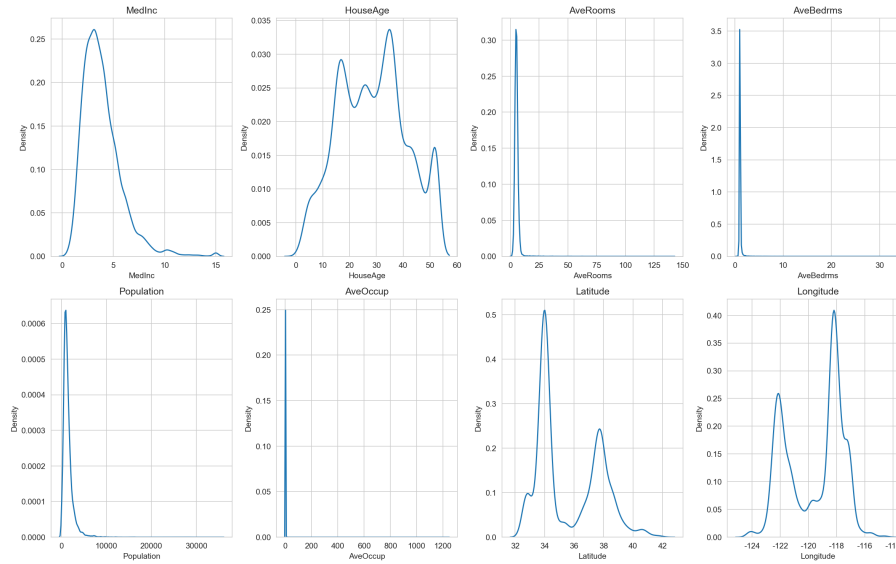


Figure 1

- KDE with Transformed Data

In Figure 1, we can see that AveRooms, AveBedrms, Population, and AveOccup are obviously right-skewed with long tails. To make the distribution clearer, we filtered out data points above the 99th percentile and then transformed these features using the $\log_{1p}$ function:

$$y = \ln(x + 1)$$

where y is the transformed feature and x is the original raw data. The KDE of the transformed data is shown in Figure 2.

KDE Distributions of All Continuous Features (California Housing)

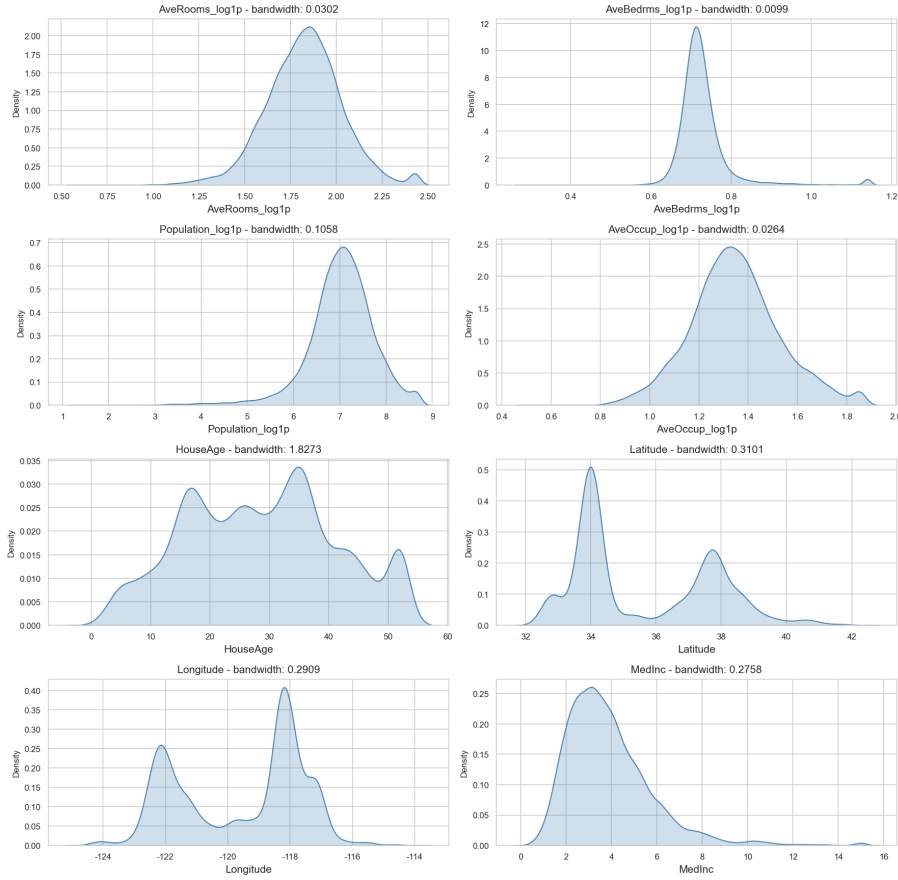


Figure 2

1.2 Method and Parameters for Kernel Width

1.2.1 Final Kernel Bandwidth

The kernel bandwidth is set using **Scott's Rule** without robust correction:

$$h = 1.059 \times \sigma \times n^{-1/5}$$

- h : optimal kernel bandwidth
- 1.059: constant for Gaussian kernel
- σ : sample standard deviation
- n : sample size

1.2.2 Process and Reason for Using Scott's Rule

When determining the kernel width, I first analyzed the relationship between the kernel width and my research goal. Since I aimed to observe clear trends in the data from the visualization—such as the type of distribution and the number of peaks—I needed a relatively smooth curve, which suggests a smaller kernel width.

I also reviewed commonly considered parameters and their roles for this dataset, including the sample size (to avoid noise while preserving real trends), the type of data distribution (using a smaller kernel width to sharpen peaks and improve interpretability), and the standard deviation.

After manually testing several values with unsatisfactory results, I realized that balancing these parameters to choose an appropriate kernel width is non-trivial. I then investigated standard, widely used methods for kernel bandwidth selection and finally adopted **Scott's method**, which is both common and effective.

I implemented a simplified version of Scott's formula through programming, rather than using the built-in function from Python libraries, because the library implementation is more complex, encapsulated, and does not provide direct access to the exact bandwidth value.

The outcome matched my expectations: the resulting curve was smooth, retained important structural features, and clearly showed significant peaks. Therefore, I used this method and its corresponding parameters to determine the kernel width. All the exact values of bandwidth are remarked in figures.

1.3 Discussion of Distribution and Interesting Phenomena

1. AveRooms (Average Rooms per Household)

- Distribution: Unimodal, strongly right-skewed, sharp peak at ~ 1.85 (transformed).
- Finding: The log1p transformation compresses the right tail. Most households have about 6–7 rooms, with a small secondary bump. I think developers tend to build homes with either many or few rooms. Large homes appeal to wealthy buyers, while small ones suit average people, making medium-sized homes less attractive. In other words, the second peak may reflect a threshold where more rooms sharply increase prices. This could also be caused by data collection methods. I have several possible explanations but am unsure which is correct.

2. AveBedrms (Average Bedrooms per Household)

- Distribution: Narrow unimodal peak at ~ 0.71 (transformed), very small right tail.
- Finding: Much more concentrated than AveRooms. Most households have 1–2 bedrooms, with a small secondary bump like AveRooms.

3. Population (Block Group Population)

- Distribution: Unimodal, right-skewed, peak at ~ 7.07 (transformed).
- Finding: After log transformation, population is concentrated around 1100–1200 people per block group, with a small secondary bump like AveRooms.

4. AveOccup (Average Household Occupancy)

- Distribution: Unimodal, slightly right-skewed, peak at ~ 1.33 (transformed).
- Finding: The peak corresponds to about 2.8 people per household, consistent with typical U.S. household sizes. A small secondary bump like AveRooms exists.

5. HouseAge (Median House Age)

- Distribution: Multi-modal, with three clear peaks at ~ 16.84 , ~ 25.85 , ~ 34.88 , plus a smaller peak near 50.
- Finding: This reveals clear cohorts of housing stock:
 - Early 20th-century homes (peak ~ 17)
 - Mid-century homes (peak ~ 26)

- Late 20th-century homes (peak ~ 35)
- Very new homes (near 0) and very old homes (near 50)

This reflects historical housing booms in California.

6. Latitude

- Distribution: Bimodal, dominant peak ~ 33.99 , secondary peak ~ 37.73 .
- Finding: Peaks correspond to Southern California (Los Angeles/Orange County, $\sim 34^\circ\text{N}$) and Northern California (Bay Area, $\sim 38^\circ\text{N}$), showing the state's two major metropolitan clusters. I got the exact placenames by asking AI.

7. Longitude

- Distribution: Bimodal, peaks at ~ -122.14 and ~ -118.18 .
- Finding: Peaks match the Bay Area ($\sim -122^\circ\text{W}$) and Southern California coast ($\sim -118^\circ\text{W}$), confirming coastal population concentration. I got the exact placenames and infomations by asking AI.

8. MedInc (Median Income)

- Distribution: Unimodal, right-skewed, peak at ~ 3.11 .
- Finding: I believe this distribution is consistent with common sense.

2. Task 2: Bimodal Quantitative Features

1. Latitude
2. Longitude

(I got the exact placenames and infomations about the Latitude and Longitude at peaks by asking AI.)

1. Latitude (peaks at ~ 32.86 , ~ 33.99 , ~ 37.73)

- The two dominant peaks correspond to California's two major population clusters:
 - **Southern California ($\sim 34^\circ\text{N}$)**: Includes Los Angeles, Orange County, and San Diego, the state's most populous region.
 - **Northern California ($\sim 38^\circ\text{N}$)**: Includes the San Francisco Bay Area (San Francisco, Oakland, San Jose), another major economic and population center.
- The smaller peak at ~ 32.86 likely reflects the San Diego metropolitan area, a distinct southern coastal cluster.

2. Longitude (peaks at ~ -122.14 , ~ -118.18)

- The peaks align with California's coastal population centers:
 - **Bay Area ($\sim -122^\circ\text{W}$)**: Centered on San Francisco and the surrounding Bay Area, a key tech and urban hub on the central coast.
 - **Southern California ($\sim -118^\circ\text{W}$)**: Centered on Los Angeles and Orange County, the state's largest urban area on the southern coast.

- This bimodality directly reflects California's population being concentrated along the Pacific coast in two distinct metropolitan regions, rather than being evenly distributed inland.
-

3. Task 3: Features Where the Average Is Not Representative

3.1 List of Features

AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude, MedInc, HouseAge

3.2 Explanations

- **Right-skewed features (AveRooms, AveBedrms, Population, AveOccup, MedInc)**

Their average isn't representative. Because the average is pulled upward by the long right tail of large values, while the majority of data points cluster around the left (lower values) near the median or peak.

- **Bimodal features (Latitude, Longitude)**

In a bimodal distribution (Latitude, Longitude), the data has two distinct and separate peaks, so the average lies in the low density space between the two clusters, representing nothing characteristic of the data.

- **Multi-modal feature (HouseAge)**

In a multimodal distribution (HouseAge), I think it is hard to say whether it is representative. But as far as I'm concerned, the average seems to lie between the peaks (even close to them), and it cannot represent any of the main groups, so I think it is not representative.

4. Task 4: Most Predictive Features for House Value

4.1 Top 3 Predictive Features

- median_income
- latitude
- longitude

4.2 Reasons

- **median_income**: Strongly correlated with house value. Higher-income areas can afford more expensive housing, and high-end housing is often built in concentrated areas.
- **latitude / longitude**: Location is one of the most important factors for housing prices. Earlier analysis showed that latitude and longitude relate to location popularity, which I believe is highly correlated with housing values.